A Comprehensive Reliability Assessment of hBN-encapsulated TMD FETs and Inverters

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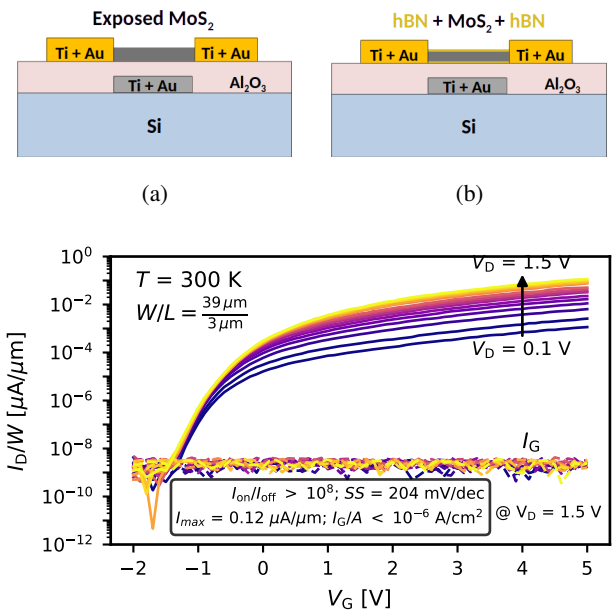
I. Introduction

II. Methods and Fabrication



(a) (b)

(c)

Fig. 1. **Cross-section schematics**: (a) exposed and (b) hBN-encapsulated MoS_2 FETs. (c) **Performance of the fabricated devices**: $I_{D(G)}$-V_G curves for different drain voltages and the main device metrics.

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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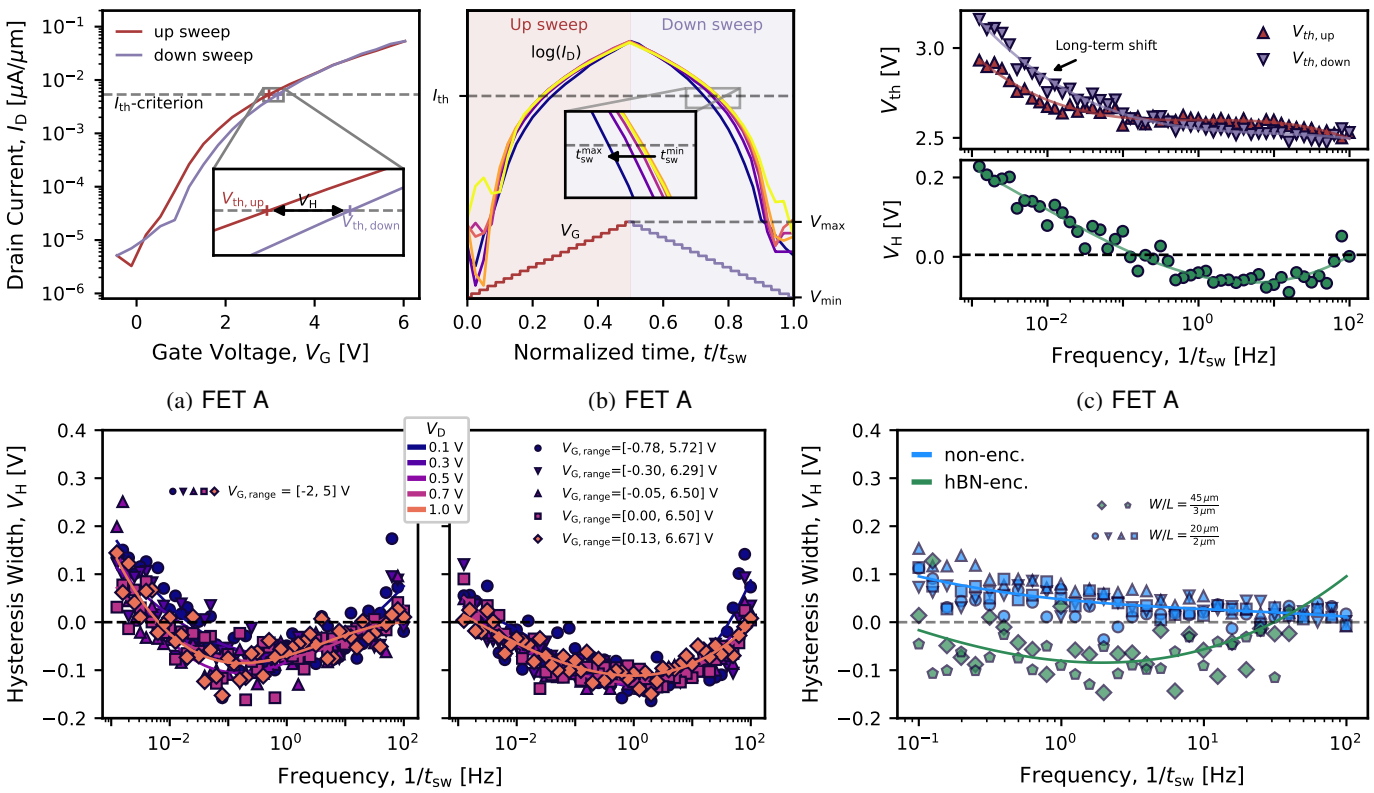
References

Please number citations consecutively within brackets [1].

References

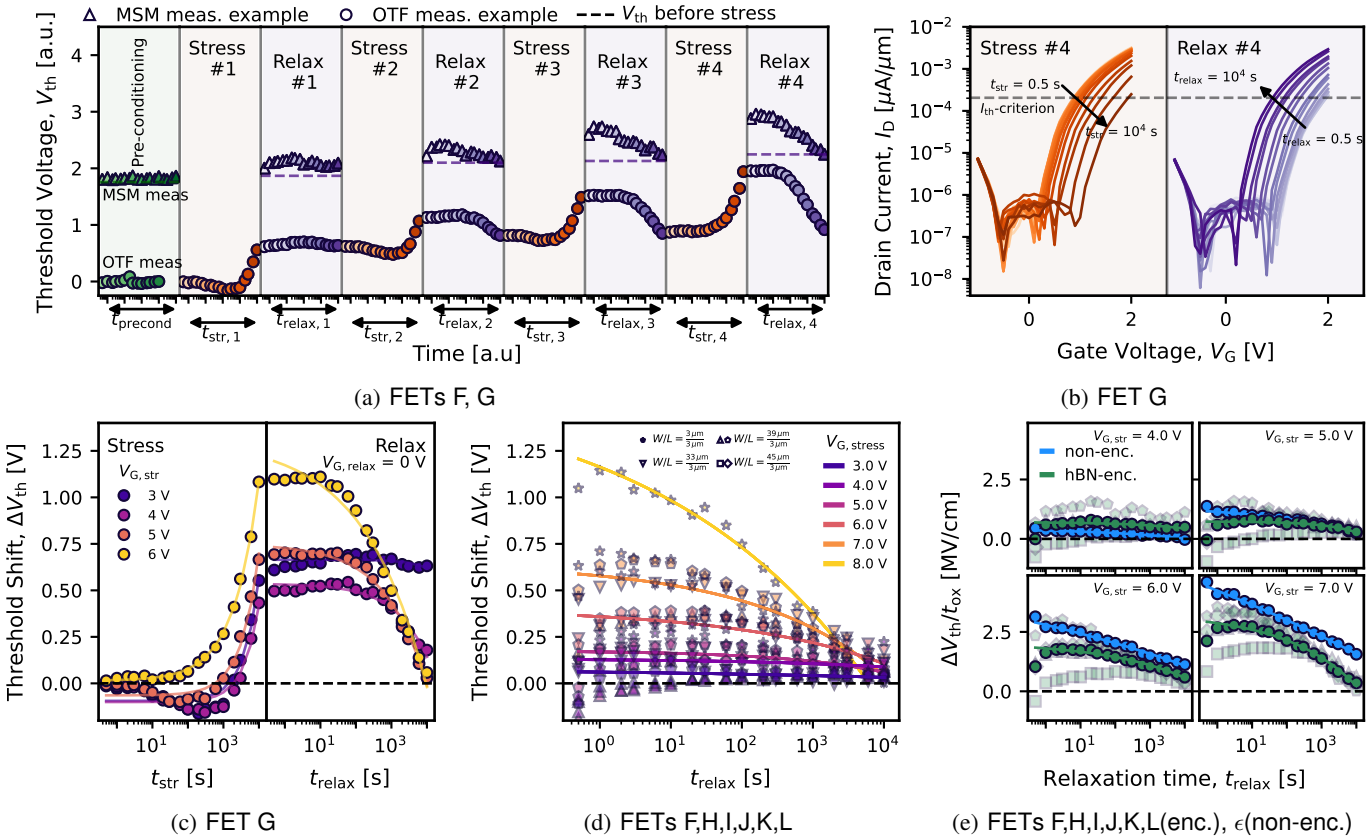
[1] J. F. Fuller, E. F. Fuchs, and K. J. Roesler. "Influence of harmonics on power distribution system protection". In: *IEEE Trans. Power Delivery* 3 (Apr. 1988), pp. 549–557.



(a) FET A (b) FET A (c) FET A

(d) FETs B, C (e) FETs D, E(enc.), $\alpha, \beta, \gamma, \delta$(non-enc.)

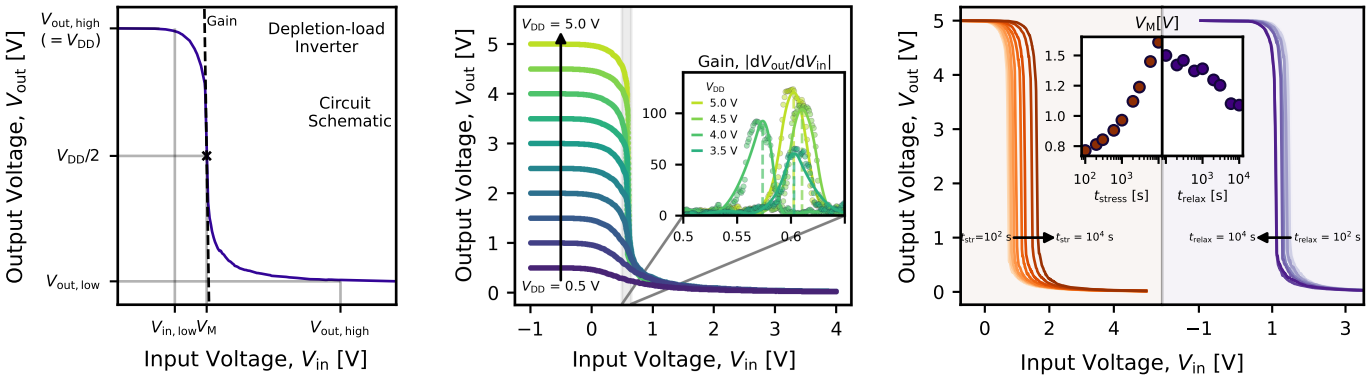
Fig. 2. (a) **Typical hysteresis curve**: the hysteresis width, V_H, is defined as the difference between the threshold voltages extracted during the up and down sweeps, i.e., $V_H = V_{th,up} - V_{th,down}$ evaluated at a constant current criterion I_{th}. (b) **Hysteresis dependence on the gate voltage sweep rate**: $\log(I_D)$ and V_G are shown as a function of the normalized sweep time, t/t_{sw}, for different sweep times, t_{sw}. (c) **Hysteresis-$1/t_{sw}$ curves**: Threshold voltages, $V_{th,up}$ and $V_{th,down}$, and hysteresis width, V_H, are shown as a function of the sweep frequency, $1/t_{sw}$. (d) **Hysteresis for varying V_D**: The hysteresis is independent from the applied drain bias, as evaluated on two different devices. More consistent hysteresis results are obtained when using an adaptive scheme for the gate voltage range targeting the same overdrive and the same sweep range in the subthreshold (right), as compared to a fixed gate voltage range (left). (e) **Hysteresis on encapsulated vs. non-encapsulated devices**: The hysteresis width on encapsulated devices is negative (counter-clockwise) and on non-encapsulated devices it was of a similar magnitude but positive (clockwise) evaluated at $V_D = 0.1V$.



(a) FETs F, G (b) FET G

(c) FET G (d) FETs F,H,I,J,K,L (e) FETs F,H,I,J,K,L(enc.), ϵ(non-enc.)

Fig. 3. (a) Example of a hysteresis curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.



(a) Inverter A (b) Inverter A (c) Inverter B

Fig. 4. (a) Example of a TDDB curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.